

1. A home uses fuel oil at the rate $r(t) = 10 + 8\sin\left(\frac{t}{60}\right)$ gallons per day, where t is the number of days from the beginning of the heating season. To the nearest gallon, what is the total amount of fuel oil used from $t = 0$ to $t = 60$ days?

1.51(2)

To find total we take the integral of $r(t)$

Ans: 821

$$\int_0^{60} (10 + 8\sin(\frac{t}{60})) dt = (10t - 480\cos(\frac{t}{60})) \Big|_0^{60} = (600 - 480\cos(1)) + (480\cos(0)) = 600 - 480\cos(1) + 480 = 600 - 480\cos(1) + 480$$

2. At time $t = 0$ minutes, a tank contains 48 gallons of water. For $0 \leq t \leq 4$ minutes, water flows into the tank at a rate of $E(t) = 12t\sin\left(\frac{\pi}{4}t\right)$ gallons per minute, and water leaks out of the tank at a rate of

$L(t) = \frac{6t^2}{t+2}$ gallons per minute. How many gallons of water are in the tank at time $t = 4$ minutes?

1/ (3)

$$48 + \int_0^4 (12t\sin(\frac{\pi}{4}t) - (\frac{6t^2}{t+2})) dt$$

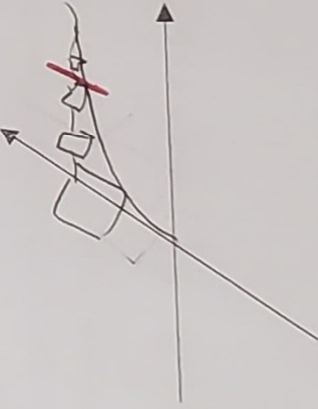
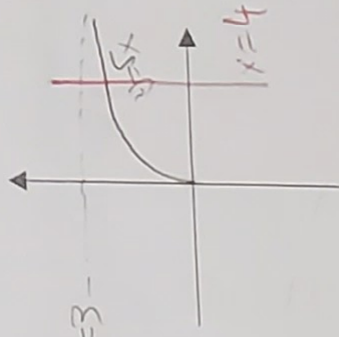
$$(-\frac{4}{\pi}\cos(\frac{\pi}{4}t) + \frac{10}{\pi^2}\sin(\frac{\pi}{4}t)) \Big|_0^4 - \int_0^4 (\frac{6t^2}{t+2}) dt$$

Ans 83 gallons

3. The base of a solid is the region in the first quadrant bounded by the y-axis, the graph of

$y = \sqrt{x}$, the y-axis, horizontal line $y = 3$ and the vertical line $x = 4$. For this solid, each cross section perpendicular to the x-axis is a square. What is the formula for the area $A(x)$ of the cross section? Draw the diagram. (2)

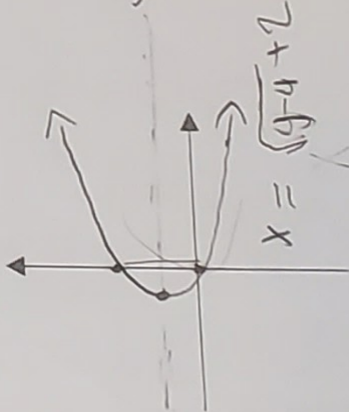
$A(x) = (3 - \sqrt{x})^2$



1.5/2

4. The region bounded by the graph of $x = 4y - y^2$ and the y-axis is the base of a solid. For this solid, each cross section perpendicular to y-axis is an equilateral triangle. What is the formula for the area $A(y)$ of the cross section? Draw the diagram.

graph is wrong side



$A(y) = \frac{\sqrt{3}}{4} (\text{base})^2 = \frac{\sqrt{3}}{4} (4 - y)^2$

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5. A particle moves along the x-axis so that its velocity at time t , $0 \leq t \leq 5$, is given by $v(t) = 3(t-1)(t-3)$. Suppose that the initial position of the particle is $x(0) = 5m$.

(1)

- a) What is the particle's position at time $t = 4s$?

$$v(t) = 3(t^2 - 4t + 3) \quad \text{not change}$$

$$\int_0^4 v(t) dt = 3 \int_0^4 (t^2 - 4t + 3) dt = 3 \left(\frac{t^3}{3} - 2t^2 + 3t \right) \Big|_0^4 = \left(\frac{64}{3} - 32 + 12 \right) + 5 = \frac{19}{3} + 5 = \frac{34}{3} \text{ m} \quad \text{Ans } 9m$$

- b) What is the position of the particle at time t sec?

$$= \int_0^t v(x) dx + 5m$$

as a polynomial

1/ (2)

- c) Find the total distance traveled by the particle

$$\int_0^5 |v(t)| dt = 5 + \int_0^1 v(t) dt + \int_1^3 v(t) dt + \int_3^5 v(t) dt = 5 + \left(\frac{4}{3} \right) + \left(\frac{20}{3} \right) = \frac{28}{3} \text{ m is total distance travelled} \quad 33m$$

0/ (1)

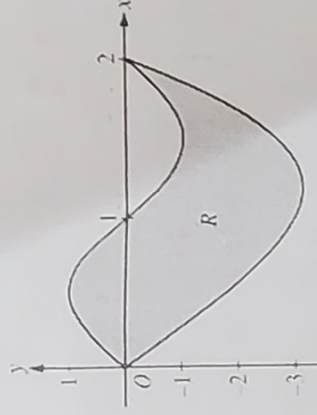
- d) Find the average velocity of the particle over the interval $0 \leq t \leq 5$.

$$\frac{\int_0^5 v(t) dt}{5s} = \frac{\left(\frac{5}{3} - 50 + 15 \right) \left(\frac{23}{3} \right)}{5} = \frac{20}{15} = \frac{4}{3} \text{ m/s is avg velocity} \quad 4 \text{ m/s}$$

6. Let R be the region bounded by the graphs of $y = \sin(\pi x)$ and $y = x^3 - 4x$, as shown below. Find area of R .

1.5 (2)

$$\int_0^2 (\sin(\pi x) - (x^3 - 4x)) dx$$



$$\int_0^2 (\sin(\pi x) + 4x - x^3) dx = \left(-\cos(\pi x) + 2x^2 - \frac{x^4}{4} \right) \Big|_0^2$$

7.

$$1 + 8 - 4 = 5$$

Total area is 5

4

What is the volume of the solid generated when the region bounded by the graph of $x = \sqrt{y-2}$ and the lines $x = 0$ and $y = 5$ is revolved about:

a) the x-axis?

2/3

X axis use washer method

$$x = 5 \quad \text{outer} \quad y^2 + 2 = 5 \quad x = \sqrt{3}$$

$$f(y) = \sqrt{y^2 + 2} + 2 \leftarrow \text{max}$$

$$\int_0^5 (25 - (x^2 + 2)^2) dy$$

$$\int_0^5 (25 - (x^4 + 4x^2 + 4)) dy$$

$$\pi \left(25x - \frac{x^5}{5} - \frac{4}{3}x^3 + 4x \right) \Big|_0^5$$

b) the y-axis?

2/3

use washer method

$$\int_2^5 ((y-2)^2) dy$$

$$\pi \int_2^5 (y-2)^2 dy$$

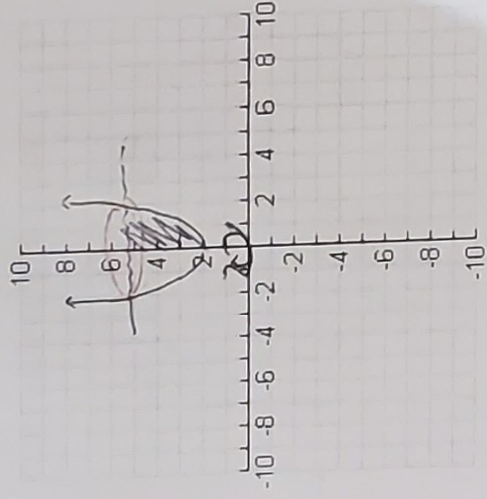
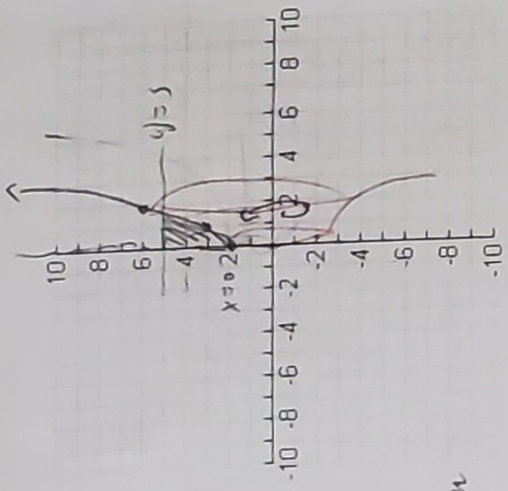
$$\pi \left(\frac{y^3}{3} - 2y^2 \right) \Big|_2^5$$

$$\frac{25}{3} - 2 \sim \pi \frac{21}{2} = \text{volume}$$

$$\left(\frac{25}{2} - 4 \right) - (2 - 4)$$

Ans $\frac{9\pi}{2}$

write properly hard to follow



9. Let R be the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$. Calculate the volume of the solid generated when R is rotated about the horizontal line $y=2$. Draw the diagram.

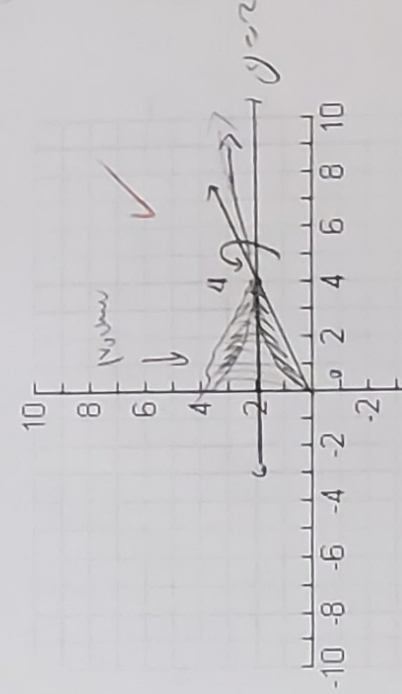
3.5/4

$$\text{inner} = 2 - \sqrt{x}$$

$$\text{outer} = 2 - \frac{x}{2}$$

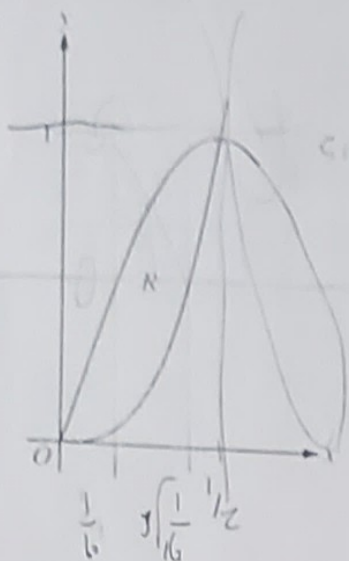
$$\pi \int_0^4 \left(\left(2 - \frac{x}{2} \right)^2 - \left(2 - \sqrt{x} \right)^2 \right) dx$$

$$\pi \int_0^4 \left(\frac{x^2}{4} - 2x + 4 \right) - (x - 4\sqrt{x} + 4) dx$$



11. Let R be the shaded region in the first quadrant enclosed by the y-axis and the graphs of

$f(x) = 8x^3$ and $g(x) = \sin(\pi x)$, as shown below.



a)

find area of R

(3)

Then intersect at $x = \frac{1}{2}$

$$\sin(\pi x) = 1$$

$$x = \frac{1}{2}$$

$$R = \int_0^{1/2} (\sin(\pi x) - 8x^3) dx$$

$$R = \left(-\frac{1}{\pi} \cos(\pi x) - 2x^4 \right) \Big|_0^{1/2}$$

$$= -\frac{2}{16} + \frac{1}{\pi} \cos(\pi) = -\frac{2}{16} - \frac{1}{\pi}$$

$$\frac{16 - 2\pi}{16\pi} = \text{area}$$

b) Write, but do not evaluate, an integral expression for the volume of solid generated when R is rotated about the horizontal line $y = 1$.

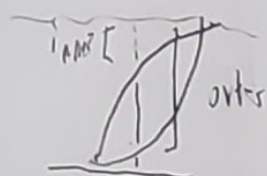
(2)

$$\pi \int_0^{1/2} \left((1 - 8x^3)^2 + (1 - \sin(\pi x))^2 \right) dx$$

axis of rotation is $y=1$

inner = $1 - \sin(\pi x)$ $\leftarrow$ so we must shift the line by subtracting

outer = $1 - 8x^3$



inner = $1 - \sin(\pi x)$
outer = $1 - 8x^3$

c) Write, but do not evaluate, an integral expression for the volume of solid generated when R is rotated about the horizontal line $y = 1$.